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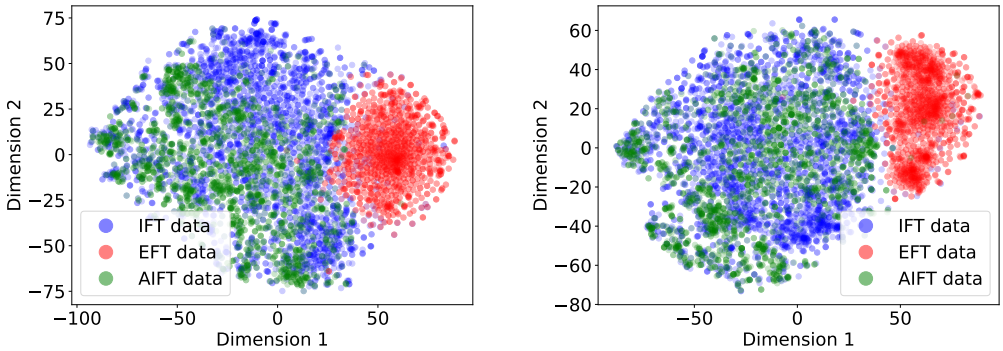
A Appendix

A.1 Distributions of IFT, EFT and AIFT data



(a) Instruction distribution of IFT, EFT and AIFT data.

(b) Response distribution of IFT, EFT, and AIFT data.

Figure 6: Distributions of both instructions and responses for IFT, EFT and AIFT data.

We have plotted the distribution of instructions for IFT, EFT and AIFT(M_1) data, and the distribution of responses for IFT, EFT and AIFT(M_1) data in Figure 6. It is clear that the IFT data and EFT data come from very different distributions while the IFT and AIFT(M_1) data come from similar distributions.

A.2 EFT Prompts

The EFT prompt which we use in our main experiments is shown in Figure 2.

Other EFT prompts we have tried At first, we took the EFT prompt from Li et al. [2024] as shown in Figure 7. However, we found that this prompt was not as effective as our additive score-counting prompt because the model needed to treat the task as a multiple-choice problem, and it was difficult for the model to break down this multiple-choice problem into sub-problems involving evaluating various aspects of the response. When using the model trained on 3,200 IFT data only, its performance on the EFT test set using our additive score-counting prompt and prompt from Li et al. [2024] is shown in Table 5.

EFT Prompt	Multiple Choice prompt	Ours
Pairwise accuracy (↑)	26.6%	65.1%
5-best % (↑)	23.5%	39.6%
Exact Match % (↑)	1.1%	10.1%
Spearman corr. (↑)	-0.18	0.25
Kendall τ corr. (↑)	-0.16	0.23

Table 5: We tried various LLM-as-Judge prompts using the model trained with 3,200 IFT data only and found that our additive score-counting prompt worked best which demonstrates significant improvements in EFT performance comparing to the prompt used by Li et al. [2024].

A.3 Self-rewarding Models Using IFT Data Only

To demonstrate the importance of the EFT data, we also trained a series of models starting with the model trained only on the IFT data. The following is the model sequence.

M_0 : Base pretrained LLM with no fine-tuning.